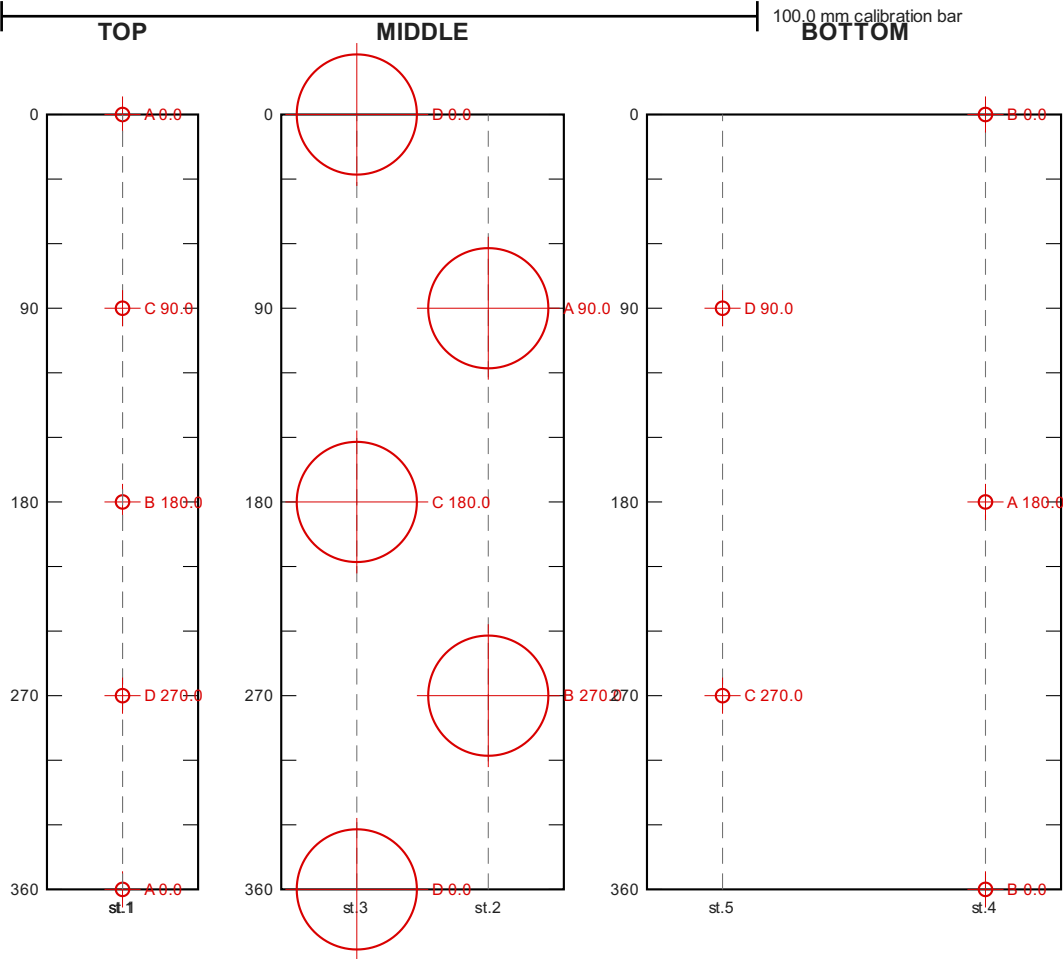


QFH mast drilling template - target 137.500 MHz

Layout inspired by the ON6JC/LW3HAZ QFH design tool by John Coppens - jcoppens.com/ant/qfh

Mast dia. 32.6 mm (circumference 102.5 mm) | support tube dia. 15.9 mm | wire dia. 1.78 mm | generated by qfh_matlab

- 1. Print at 100% / "Actual size" (never "fit to page"). Check: the calibration bar below must measure exactly 100.0 mm.
- 2. Draw a straight SEAM line along the mast. On it, mark the depth of every station (table below), measured from the antenna top plane (depth 0 = feed level).
- 3. Cut out each strip and wrap it around the mast, printed side out: 0-deg edge on the seam, station line(s) on the matching depth mark(s). The strip ends must meet on the seam.
- 4. WRAP DIRECTION MATTERS: seen from the TOP of the mast, angles must increase counter-clockwise. Wrapping the other way mirrors the antenna (LHCP instead of RHCP!).
- 5. SANITY CHECK after wrapping: deeper station lines must sit LOWER on the mast (large-loop middle below small-loop middle). If inverted, the strip is wrapped the wrong way.
- 6. Drill through the center of every circle: small circles = wire holes (dia. 1.78 mm), large circles = support-tube holes (dia. 15.9 mm).
- 7. Equal letters = same helix arm (A/B = small loop, C/D = large loop).



st.	loop	position	depth [mm]	hole dia. [mm]	angles [deg]	arms
1	small	top	0.0	1.78	0.0 / 180.0	A / B
1	large	top	0.0	1.78	90.0 / 270.0	C / D
2	small	middle	336.5	15.90	90.0 / 270.0	A / B
3	large	middle	353.9	15.90	180.0 / 0.0	C / D
4	small	bottom	672.9	1.78	180.0 / 0.0	A / B
5	large	bottom	707.7	1.78	270.0 / 90.0	C / D

Depths are to HOLE CENTERS, measured from the antenna top plane. Angles: 0 deg = seam line, increasing counter-clockwise seen from the top.